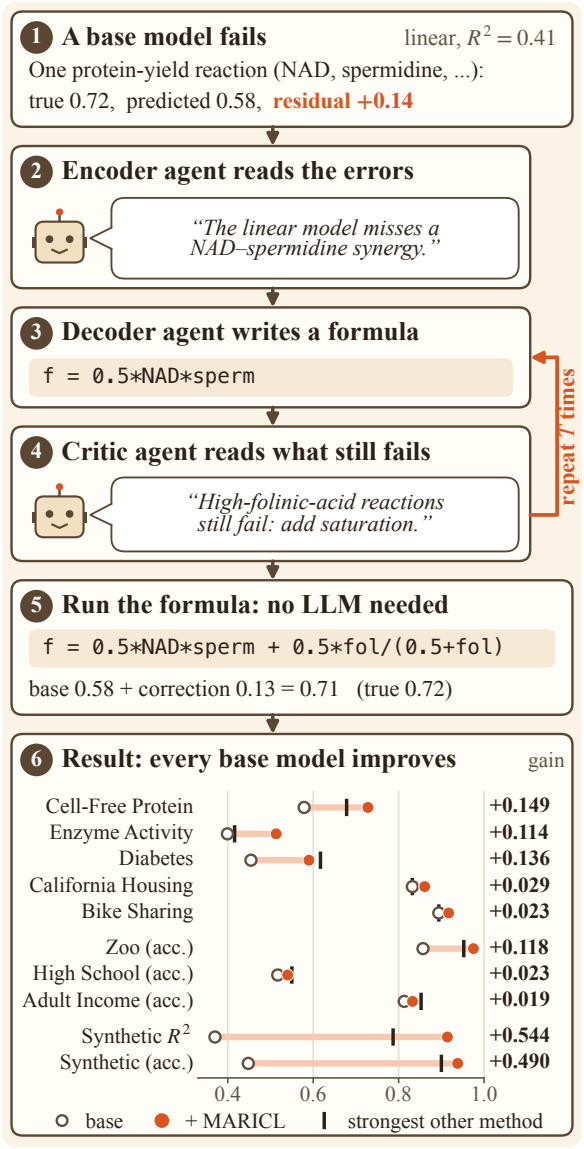

1 A base model fails
linear, $R^2 = 0.41$
One protein-yield reaction (NAD, spermidine, ...):
true 0.72, predicted 0.58, residual +0.14
2 Encoder agent reads the errors
"The linear model misses a NAD–spermidine synergy."
3 Decoder agent writes a formula
f = 0.5*NAD*sperm
repeat T times
4 Critic agent reads what still fails
"High-folinic-acid reactions still fail: add saturation."
5 Run the formula: no LLM needed
f = 0.5*NAD*sperm + 0.5*fol/(0.5+fol)
base 0.58 + correction 0.13 = 0.71 (true 0.72)
6 Result: every base model improves
gain
Cell-Free Protein +0.149
Enzyme Activity +0.114
Diabetes +0.136
California Housing +0.029
Bike Sharing +0.023
Zoo (acc.) +0.118
High School (acc.) +0.023
Adult Income (acc.) +0.019
Synthetic R^2 +0.544
Synthetic (acc.) +0.490
0.4
0.6
0.8
1.0
base
+ MARICL
strongest other method